310407/310907

Roll No

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B. Tech. III - Sem. (Main) Exam., (Academic Session 2021- 2022)

Computer Science and Engineering

3CS4 – 07 Software Engineering

Common for CSE/IT

Time: 2½ Hours **Maximum Marks: 120**

Min. Passing Marks:

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 6 × 3 marks = 18 marks. Candidates have to answer **six** questions out of **ten***

Part – B: *Analytical/Problem solving questions 3 × 10 marks = 30 marks. Candidates have to answer **three** questions out of **seven**.*

Part – C: *Descriptive/Analytical/Problem Solving questions 3 ×24 marks = 72 marks. Candidates have to answer **three** questions out of **five**.*

*Schematic diagrams must be shown wherever necessary. **Any data you feel missing** may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.*

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL

2. NIL

PART – A

Q.1 Define SDLC with its various phases.

Q.2 Differentiate between verification and validation.

Q.3 What do you mean by metrics?

Q.4 Discuss in brief about Project Scheduling.

Q.5 Distinguish between DFD and CFD.

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Q.6 Define SRS. What are the characteristics of SRS?

Q.7 Differentiate between class diagram and use case diagram.

Q.8 What do you mean by UML? What are the three types of modeling in UML?

Q.9 Discuss design documentation in software design.

Q.10 Distinguish between architectural and procedural design.

PART – B

Q.1 What is Prototyping? How it is useful for the development of requirement specification? Explain with Prototyping methods and tools.

Q.2 Explain iterative waterfall model and spiral model for software life cycle and discuss various activities in each phase. https://www.btubikaner.com

Q.3 Describe LOC and FP estimation techniques for software project estimation.

Q.4 Describe object-oriented analysis process. What is CRC modeling? Explain in detail.

Q.5 Define finite state machine (FSM) model. Draw a FSM model for the following description:

$\Gamma = \{S, SI\}$, $\Sigma = \{a, b\}$, S= starting state, $\delta(s, a) = SI$, $\delta(SI, b) = S$

Q.6 What are the differences between structural and object oriented analysis? Explain in detail.

Q.7 List software design fundamentals and explain each by giving suitable examples.

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PART – C

Q.1 Define various design approaches in OOD. Explain different types of object oriented design methods with it's neat diagrams.

Q.2 List software design process steps. Describe effective modular design criterias to design the effective modularity with suitable example.

Q.3 What are the purposes of Data flow diagrams, control flow diagrams and Entity- relationship diagrams? Give an example diagram of each.

Q.4 What is COCOMO? Differentiate between COCOMO-I and COCOMO-II. Suppose that a project was estimated to be 400 KLOC, calculate the effort and development time for each of the three modes.

Q.5 What is risk analysis? Explain different categories of risk analysis. Describe the difference between 'Known risks' and 'Predictable risks'.

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